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Inverse Problems and Imaging (02624)

Week 4

Tikhonov regularization: Optimization

Compact operator $K \in B(X, Y)$; X, Y are HS.

Equation with perturbation (noise) in data

$$Kx \approx y^\delta = y + d, \quad \|d\| < \delta.$$

Do not solve least squares problem

$$x_0 = \operatorname{argmin}_{x \in X} \|Kx - y^\delta\|^2.$$

Instead, for regularization parameter $\alpha > 0$ consider

$$x_\alpha = \operatorname{argmin}_{x \in X} \|Kx - y^\delta\|^2 + \alpha \|x\|^2.$$

Well-posed problem for any $y_\delta \in Y$ with unique solution

$$x_\alpha = R_\alpha y^\delta$$

Tikhonov regularization: Normal equation

The least squares optimization problem

$$\operatorname{argmin}_{x \in X} \|Kx - y^\delta\|^2$$

leads to normal equation and pseudo-inverse

$$(K^*K)x = K^*y^\delta \quad \Leftrightarrow \quad x = (K^*K)^{-1}K^*y^\delta = K^\dagger y^\delta.$$

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The Tikhonov problem

$$\operatorname{argmin}_{x \in X} (\|Kx - y^\delta\|^2 + \alpha\|x\|^2)$$

leads to the normal equation and regularized inverse

$$(K^*K + \alpha)x_\alpha = K^*y^\delta \quad \Leftrightarrow \quad x_\alpha = (K^*K + \alpha)^{-1}K^*y^\delta = R_\alpha y^\delta.$$

Tikhonov regularization: Filter factors

Formal solution to $Kx = y^\delta$ in the SVD system $\{x_j\}, \{y_j\}, \{\mu_j\}$

$$x = \sum_j \frac{(y^\delta, y_j)}{\mu_j} x_j.$$

Regularization via filtered versions

$$x_\alpha = \sum_j q_\alpha(\mu_j) \frac{(y^\delta, y_j)}{\mu_j} x_j.$$

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$$\text{TSVD} \quad x_\alpha = \sum_{\mu_j^2 > \alpha} \frac{(y^\delta, y_j)}{\mu_j} x_j, \quad q_\alpha(\mu) = \begin{cases} 1, & \mu^2 > \alpha, \\ 0, & \mu^2 \leq \alpha. \end{cases}$$

$$\text{Tikhonov} \quad x_\alpha = \sum_j \frac{\mu_j^2}{\mu_j^2 + \alpha} \frac{(y^\delta, y_j)}{\mu_j} x_j, \quad q_\alpha(\mu) = \frac{\mu^2}{\mu^2 + \alpha}.$$

Numerical implementation

Rewrite for $K = A \in \mathbb{R}^{n \times m}$ and $y \in \mathbb{R}^n$

$$\|Ax - y\|^2 + \alpha\|x\|^2 = \left\| \begin{bmatrix} A \\ \sqrt{\alpha}I \end{bmatrix} x - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|^2.$$

Implement by defining augmented matrix and solving least-squares problem.

Tikhonov as regularization strategy

Theorem

Tikhonov regularization is a regularization strategy with admissible parameter choice rule $\alpha(\delta)$ satisfying

$$\lim_{\delta \rightarrow 0} \alpha(\delta) = 0,$$
$$\lim_{\delta \rightarrow 0} \frac{\delta^2}{\alpha(\delta)} = 0.$$